

Part 11: Markov Models

Figure 1 below depicts a stochastic process which moves between three *states* (the circles) labeled 0, 1, and 2. The arrows each have probabilities associated with them. For instance, if the process is currently in state “0,” it moves to “1” with probability 0.4 on the next step, and stays at “0” with probability 0.6.

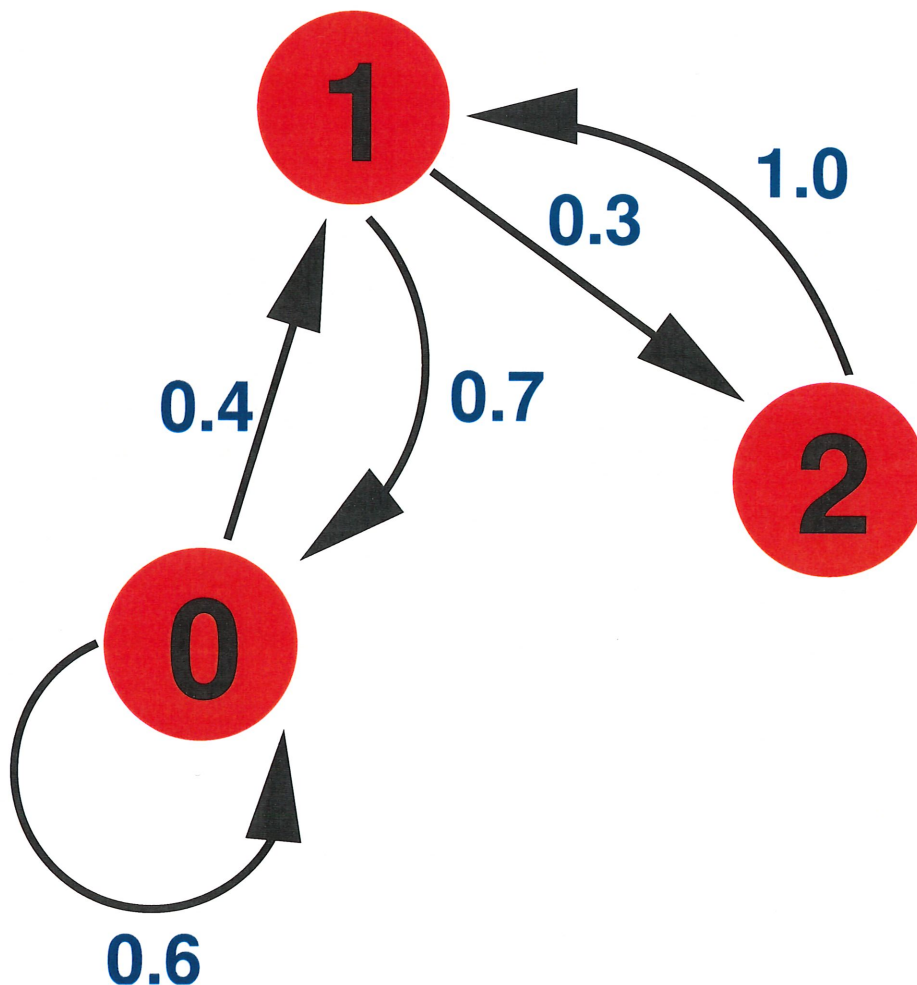


Figure 1: A transition diagram.

1 **Exercise:** Give examples of questions we could ask (and hopefully
2 answer) regarding the behavior of this process.

3 ① What is the long run prop. of time
4 spent in state 0?

5 ② If start at state 0, what is expected
6 # of steps before return to 0?
7

8 ③ What is the expected # of visits to
9 state 1 before returning to state 0?
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The Markov Chain

Recall our initial definition of a time series model as a *stochastic process* $\{X_t: t \in \mathcal{T}\}$ where $\mathcal{T}$ indexes time. Call the set of possible values for each X_t be the *state space*, denoted $\mathbb{S}$.

If $\mathcal{T} = \{0, 1, 2, \dots\}$ and $\mathbb{S}$ is discrete, then the stochastic process is a *discrete time chain*. $\mathbb{S}$ is the class of possible *states* of the chain: If it is now time n , then:

- $X_0, X_1, \dots, X_{n-1}$ are the states of the chain in the past.
- X_n is the current state of the chain.
- $X_{n+1}, X_{n+2}, \dots$ are the future states of the chain.

For the sake of convenience, we will often label the states (the elements of $\mathbb{S}$) as $0, 1, 2, \dots$.

The general question: What do the values of the random variables in the past and present tell us about the distribution of X_{n+1} ?

1 If we assume that $X_0, X_1, \dots$ are independent, then

$$2 \quad P(X_{n+1} = j | X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) = P(X_{n+1} = j).$$

3 In other words, we assume the past gives us no information. Al-
4 though this is very easy to work with from a theoretical and compu-
5 tational point of view, it is not realistic in many practical situations.

6 If we assume that the distribution of X_{n+1} is dependent on **all** of the
7 past $(X_0, X_1, \dots, X_n)$, the model will be cumbersome and impracti-
8 cal.

9 The middle ground: A *Markov chain of order k* assumes that

$$10 \quad P(X_{n+1} = j | X_0 = i_0, X_1 = i_1, \dots, X_{n-1} = i_{n-1}, X_n = i) =$$

$$11 \quad P(X_{n+1} = j | X_{n-k+1} = i_{n-k+1}, \dots, X_{n-1} = i_{n-1}, X_n = i).$$

12 So, **conditional on** $X_{n-k+1}, X_{n-k+2}, \dots, X_n$, the state X_{n+1} is indepen-
13 dent of $X_0, X_1, \dots, X_{n-k}$.

We focus on *first order Markov chains*, i.e. Markov chains of order $k = 1$. In fact, it is possible to make any Markov chain of any order into a first order chain by enlarging the state space appropriately.

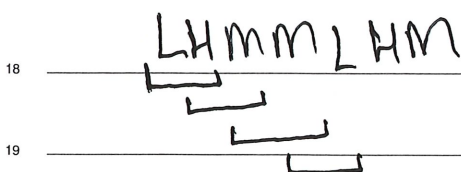
Exercise: I believe that a model with three volatility *regimes* provides a realistic view of market conditions:

$$\mathbb{S} = \{\text{Low, Medium, High}\}.$$

I believe that the distribution over $\mathbb{S}$ for tomorrow's volatility can be adequately modelled by the state yesterday and today. Thus, this is a second order Markov chain. Show that by enlarging the state space, you can create a new process which will have the same behavior as this second order chain, but is a first order Markov chain.

$$\mathbb{S}' = \{LL, LM, LH, \dots, HH\}$$

If currently in state LM, can only move to ML, MM, MH



1 For a first order Markov chain,

$$\begin{aligned} 2 \quad P(X_{n+1} = j | X_0 = i_0, X_1 = i_1, \dots, X_{n-1} = i_{n-1}, X_n = i) = \\ 3 \quad P(X_{n+1} = j | X_n = i). \end{aligned}$$

4 Since this is so important, we denote $P_{ij}(n) = P(X_{n+1} = j | X_n = i)$.

5 These are the *transition probabilities at time n*. In other words, if at time
6 n we are in state i , then $P_{ij}(n)$ is the probability the next “step” of the
7 chain is to state j .

8 There is yet another simplification that is common: We assume that
9 the chain is *time homogeneous* if we assume that $P_{ij}(n)$ does not de-
10 pend on n ; in this case we just denote the transition probability from
11 state i to state j as P_{ij} .

12 Time homogeneous, first order chains are prevalent, and are often
13 referred to as simply *Markov chains*.

14 **Exercise:** $\{X_n : n \geq 0\}$ is a (time homogeneous, first order) Markov
15 chain. Are the random variables X_1 and X_3 independent?

16 NO. Conditional on X_2 , X_1 and X_3 are independent

$$\begin{aligned} 17 \quad P(X_3 = j | X_1 = i_1, X_2 = i_2) &= P(X_3 = j | X_2 = i_2) \\ 18 \quad P(X_3 = j | X_1 = i_1) &\neq P(X_3 = j) \end{aligned}$$

The Transition Matrix

It is natural to place the transition probabilities for a Markov chain in a matrix. Call this the **transition matrix**, denoted $\mathbf{P}$.

$$\mathbf{P} = \begin{bmatrix} P_{00} & P_{01} & P_{02} & \cdots \\ P_{10} & P_{11} & P_{12} & \cdots \\ P_{20} & P_{21} & P_{22} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Each row of $\mathbf{P}$ must sum to one, but this is not true of the columns.

Exercise: Write out the transition matrix that corresponds to the diagram at the start of this Part.

$$\mathbf{P} = \begin{bmatrix} 0.6 & 0.4 & 0 \\ 0.7 & 0 & 0.3 \\ 0 & 1 & 0 \end{bmatrix}$$

A Markov chain is called *finite* if $\mathcal{S}$ consists of a finite number of states.

1 **Example: Credit Ratings Migration**

2 The following pages are taken from Hull, *Options, Futures, and Other*
3 *Derivatives*, Fifth Edition.

4 Table 26.8 depicted is a transition matrix for a Markov chain model
5 with nine states. The state space consists of the different possible
6 bond ratings, and a time step is a year. Such a model is a useful
7 approximation to the way in which a firm's bond ratings changes
8 from one year to the next.

26.7 CREDIT RATINGS MIGRATION

Over time, bonds are liable to move from one rating category to another. This is sometimes referred to as *credit ratings migration*. Rating agencies produce from historical data a *ratings transition matrix*. This shows the percentage probability of a bond moving from one rating to another during a certain period of time. Usually the period of time is one year. Table 26.8 shows data produced by S&P. It shows that a bond that starts with a BBB credit rating has an 89.24% chance of still being a BBB at the end of one year. It has a 0.24% chance of defaulting during the year, a 4.44% chance of dropping to BB, and so on. Appendix 26A shows how a table such as Table 26.8 can be used to calculate transition probabilities for time periods other than one year.

The probabilities in Table 26.8 are based on historical data and are therefore real-world probabilities. As we will see later, they are useful for scenario analysis and the calculation of value at risk. For valuing derivatives that depend on credit rating changes, we require a risk-neutral transition matrix. This proves to be quite difficult to estimate in practice.¹⁵ One approach is based on bond prices. To provide a simple example, we suppose that there are three rating categories, A, B, and C, and that D is used to denote default. Table 26.9 shows hypothetical data on the cumulative risk-neutral probabilities of default. This could be calculated from bond prices as described in Section 26.1.

Suppose that M is a 4×4 matrix showing the risk-neutral transition probabilities. This means that M_{11} is the probability that an A-rated company stays A-rated; M_{23} is the probability that a B-rated becomes C-rated; M_{34} is the probability that a C-rated company defaults; and so on. Define d_i as a vector that is the i th column of the matrix in Table 26.9. It follows that $d_2 = Md_1$, $d_3 = Md_2 = M^2d_1$, $d_4 = Md_3 = M^3d_1$, and $d_5 = Md_4 = M^4d_1$. There are nine unknown elements of M . These are M_{ij} for $1 \leq i, j \leq 3$.¹⁶ Table 26.9 defines 15 relationships that should hold. (These are the probabilities of A-, B-, and C-rated bonds defaulting within 1, 2, 3, 4, and 5 years.)

Table 26.8 One-year transition matrix of percentage probabilities

Initial rating	Rating at year-end							
	AAA	AA	A	BBB	BB	B	CCC	Default
AAA	93.66	5.83	0.40	0.09	0.03	0.00	0.00	0.00
AA	0.66	91.72	6.94	0.49	0.06	0.09	0.02	0.01
A	0.07	2.25	91.76	5.18	0.49	0.20	0.01	0.04
BBB	0.03	0.26	4.83	89.24	4.44	0.81	0.16	0.24
BB	0.03	0.06	0.44	6.66	83.23	7.46	1.05	1.08
B	0.00	0.10	0.32	0.46	5.72	83.62	3.84	5.94
CCC	0.15	0.00	0.29	0.88	1.91	10.28	61.23	25.26
Default	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00

Source: Standard & Poor's, January 2001.

¹⁵ Fortunately relatively few credit derivatives have payoffs dependent on rating changes.

¹⁶ Note that $M_{i,4} = 1 - M_{i,1} - M_{i,2} - M_{i,3}$, and $M_{4,j} = 1$ when $j = 4$ and zero otherwise.

Table 26.9 Cumulative percentage probabilities of default for rating categories

<i>Initial rating</i>	<i>1 year</i>	<i>2 year</i>	<i>3 year</i>	<i>4 year</i>	<i>5 year</i>
A	0.67	1.33	1.99	2.64	3.29
B	1.66	3.29	4.91	6.50	8.08
C	3.29	6.50	9.63	12.69	15.67
D	100.00	100.00	100.00	100.00	100.00

We therefore choose the nine parameters to minimize the 15 sums of the squares of the differences between the elements of $M^{i-1}d_1$ and the corresponding elements of d_i ($1 \leq i \leq 5$) in Table 26.9. The results are shown in Table 26.10. For this example the method works well. The transition probabilities are plausible and the cumulative default probabilities calculated from them are close to those in Table 26.9 (see Problem 26.13).

In practice, data on cumulative default probabilities are not always as well behaved as the data presented in Table 26.9. This has led some researchers to suggest other methods that are based on assuming that the real-world transition matrix changes in a particular way when we move from the real world to the risk-neutral world.¹⁷

Table 26.10 Best-fit risk-neutral transition matrix. (Probabilities expressed as percentages.)

<i>Initial rating</i>	<i>Rating at year-end</i>			
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
A	98.4	0.9	0.0	0.7
B	0.5	97.1	0.7	1.7
C	0.0	0.0	96.7	3.3
Default	0.0	0.0	0.0	100.0

26.8 DEFAULT CORRELATIONS

The term *default correlation* is used to describe the tendency for two companies to default at about the same time. There are a number of reasons why default correlations exist. Two companies in the same industry or the same geographic region tend to be affected similarly by external events and as a result may experience financial difficulties at the same time. Also, economic conditions generally cause average default rates to be higher in some years than in others.

¹⁷ See R. A. Jarrow, D. Lando, and S. M. Turnbull, "A Markov Model for the Term Structure of Credit Spreads," *Review of Financial Studies*, 10 (1997), 481–523; M. Kijima, "A Markov Chain Model for Valuing Credit Derivatives," *Journal of Derivatives*, 6, no. 1 (Fall 1998), 97–108.

Chapman-Kolmogorov Equations

Define the n -step transition probabilities for a Markov chain as

$$P_{ij}^{(n)} = P(X_{n+k} = j | X_k = i).$$

In other words, $P_{ij}^{(n)}$ is the probability the chain is at state j exactly n steps after being in state i . Note that because of time homogeneity of the chain, this probability does not depend on k .

The n -step transition probabilities can also be placed into a matrix, call it $\mathbf{P}^{(n)}$. By convention, we set $\mathbf{P}^{(0)} = \mathbf{I}$.

Exercise: What is $P_{12}^{(2)}$ for the Markov chain with transition matrix

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 0.2 & 0.1 & 0.7 \\ 0.8 & 0.2 & 0 \\ 0.1 & 0.1 & 0.8 \end{bmatrix} \end{matrix}$$

$$P_{12}^{(2)} = P(X_2 = 2 | X_0 = 1) = \sum_{k=0}^2 P(X_2 = 2 | X_0 = 1, X_1 = k) P(X_1 = k | X_0 = 1)$$

$$= \sum_{k=0}^2 P(X_2 = 2 | X_1 = k) P(X_1 = k | X_0 = 1)$$

$$= \sum_{k=0}^2 P_{k2} P_{1k}$$

= inner product between row "1" and col "2" of $\mathbf{P}$

1 The previous exercise hints that the matrix $P^2 = PP$ consists of the
2 two-step transition probabilities.

3 This can be extended: In general, the entries of the matrix P^n are
4 the n -step transition probabilities. We can say that $P^{(n)} = P^n$ for all
5 $n \geq 0$. This relationship is referred to as the *Chapman-Kolmogorov*
6 *equations*.

7 **Exercise:** Explain how the matrix $P^{(n)}$ relates to *forecasting* with the
8 Markov chain model.

9 Each row of $P^{(n)}$ tells me the dist.
10 over S when looking n steps
11 into the future

12
13 E.g., could raise the credit migration
14 matrix to n^{th} power to forecast
15 the dist. over the different states
16
17
18
19